50.007 Machine Learning

Lecture 1

Introduction

Who are we?

Instructors

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Weeks 1-3 and Week 12

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Instructor
Weeks 4 – 8.5

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Outline

- Administrative details
- What is machine learning?
- Types of machine learning
- A case study for supervised learning
- Linear Classification

Administrative details

- **Course material:**
  - No required textbook
  - Recommended reading (from books or research papers) posted on eDimension
  - Lecture notes and slides: posted on eDimension
  - **Lecture slides are provided to help better understand the lecture notes and by no means should be considered as a replacement.**
  - For a better understanding of the course, please go through the lecture notes prior to each lecture.

- **Pre-requisite:**
  - Linear Algebra **(Tutorial in Week 2)**
  - Probability/statistics **(Tutorial in Week 8)**
  - Knowledge of Algorithms
  - Python Programming

Evaluation

- Homework (28%)
  - Programming and theory
  - Honour Code
    - Form study groups to work on homework
    - You can discuss with other classmates
    - Write-up solutions on your own
    - List names of anyone you talked to
- Project (20%)
- Midterm Exam (25%)
- Final Exam (25%)
- Participation (2%)

Course Goals

- Curious to discover more
- Confident of doing it yourself
- Contemplative of the theory
- Cautious of the danger

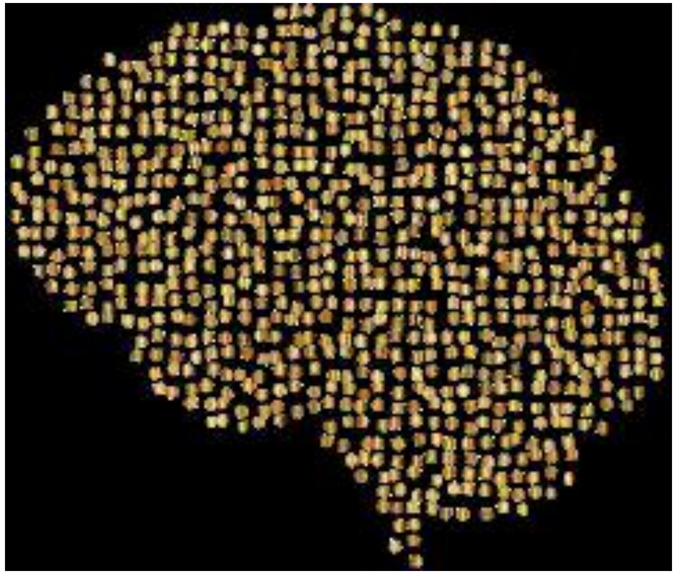
Acknowledgements

- MIT 6.036 Introduction to Machine Learning
- Stanford CS229 Machine Learning
- McGill COMP-652 Machine Learning

Introduction

What is Machine Learning?


Hardcoded


Trained

- Giving computers the **ability to learn** without being explicitly programmed – Arthur Samuel (1959)

What is Machine Learning?

Task

Performance

Experience

- Algorithms that improve their **performance** at some **task** with **experience** – Tom Mitchell (1998)

What is Machine Learning?

- A branch of **artificial intelligence**, concerned with the design and development of algorithms that allow computers to evolve behaviours based on empirical data.

- As intelligence requires knowledge, it is necessary for the computers to first **acquire knowledge through learning**.

- Machine learning is programming computers to optimize a **performance criterion** using example data or **past experience**.

Very brief history

- Studied ever since computers were invented (e.g. Samuel's checkers player)
- Very active in 1960s (neural networks)
- Died down in the 1970s
- Revival in early 1980s (decision trees, backpropagation, temporal difference learning) - coined as "machine learning"
- Exploded since the 1990s
- Now: very active research field, several yearly conferences (e.g., ICML, NIPS), major journals (e.g., Machine Learning, Journal of Machine Learning Research), rapidly growing number of researchers
- The time is right to study in the field!
  - Lots of recent progress in algorithms and theory
  - Flood of data to be analyzed
  - Computational power is available
  - Growing demand for industrial applications

Why study Machine Learning?

Engineering reasons:

- Easier to build a learning system than to hand-code a working program!
  - Robot that learns a map of the environment by exploring
  - Programs that learn to play games by playing against themselves
- Improving on existing programs,
  - Instruction scheduling and register allocation in compilers
  - Combinatorial optimization problems
- Solving tasks that require a system to be adaptive
  - Speech and handwriting recognition
  - “Intelligent” user interfaces

Why study Machine Learning?

Scientific reasons:

- Discover knowledge and patterns in high dimensional, complex data
  - Sky surveys
  - High-energy physics data
  - Sequence analysis in bioinformatics
  - Social network analysis
  - Ecosystem analysis
- Understanding animal and human learning
  - How do we learn language?
  - How do we recognize faces?
- Creating real AI!

  "If an expert system–brilliantly designed, engineered and implemented– cannot learn not to repeat its mistakes, it is not as intelligent as a worm or a sea anemone or a kitten." (Oliver Selfridge).

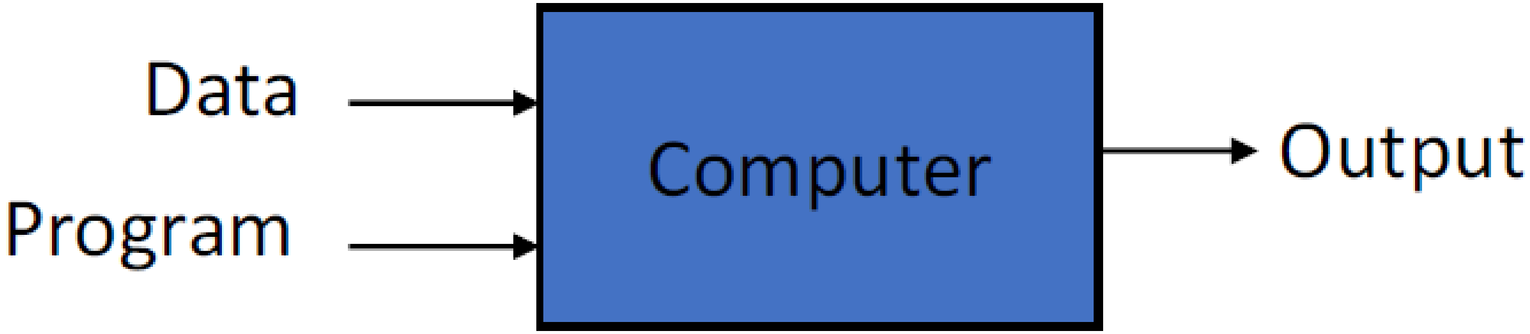
What are good Machine Learning tasks?

- There is no human expert
  E.g., DNA analysis
- Humans can perform the task but cannot explain how
  E.g., character recognition
- Desired function changes frequently
  E.g., predicting stock prices based on recent trading data
- Each user needs a customized function
  E.g., news filtering

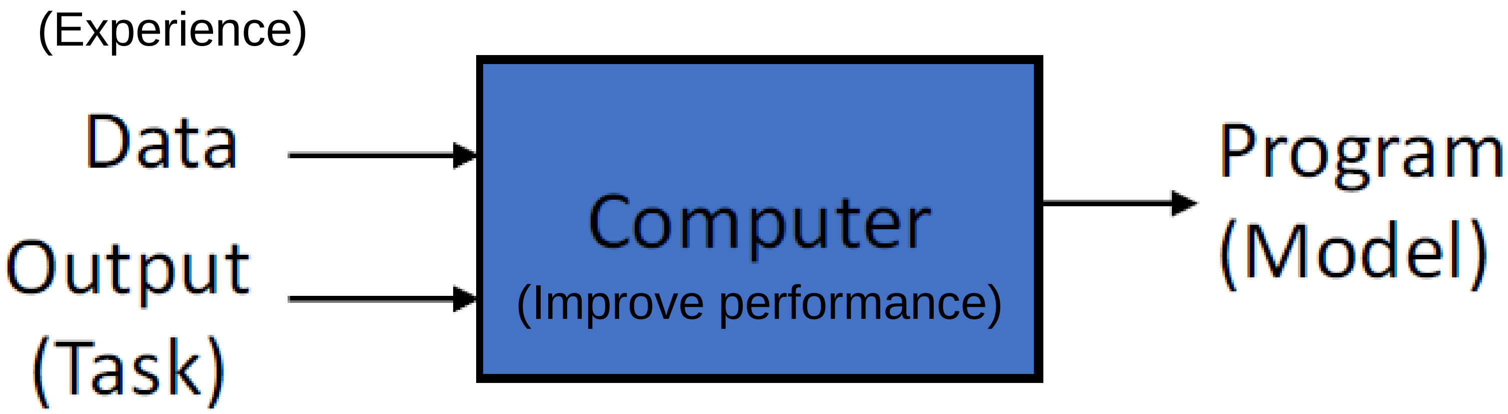
Important application areas

- **Bioinformatics:** sequence alignment, analyzing microarray data, information integration, ...
- **Computer vision:** object recognition, tracking, segmentation, active vision, …
- **Robotics:** state estimation, map building, decision making
- **Graphics:** building realistic simulations
- **Speech:** recognition, speaker identification
- **Financial analysis:** option pricing, portfolio allocation
- **E-commerce:** automated trading agents, data mining, spam, ...
- **Medicine:** diagnosis, treatment, drug design,...
- **Computer games:** building adaptive opponents
- **Multimedia:** retrieval across diverse databases

Traditional Programming



Machine Learning



Machine Learning model

- We have a model which we can use to predict new data
- E.g. Image classification

Input

model →

Dog
Building
Cat ✓
Human
Car

Machine Learning model

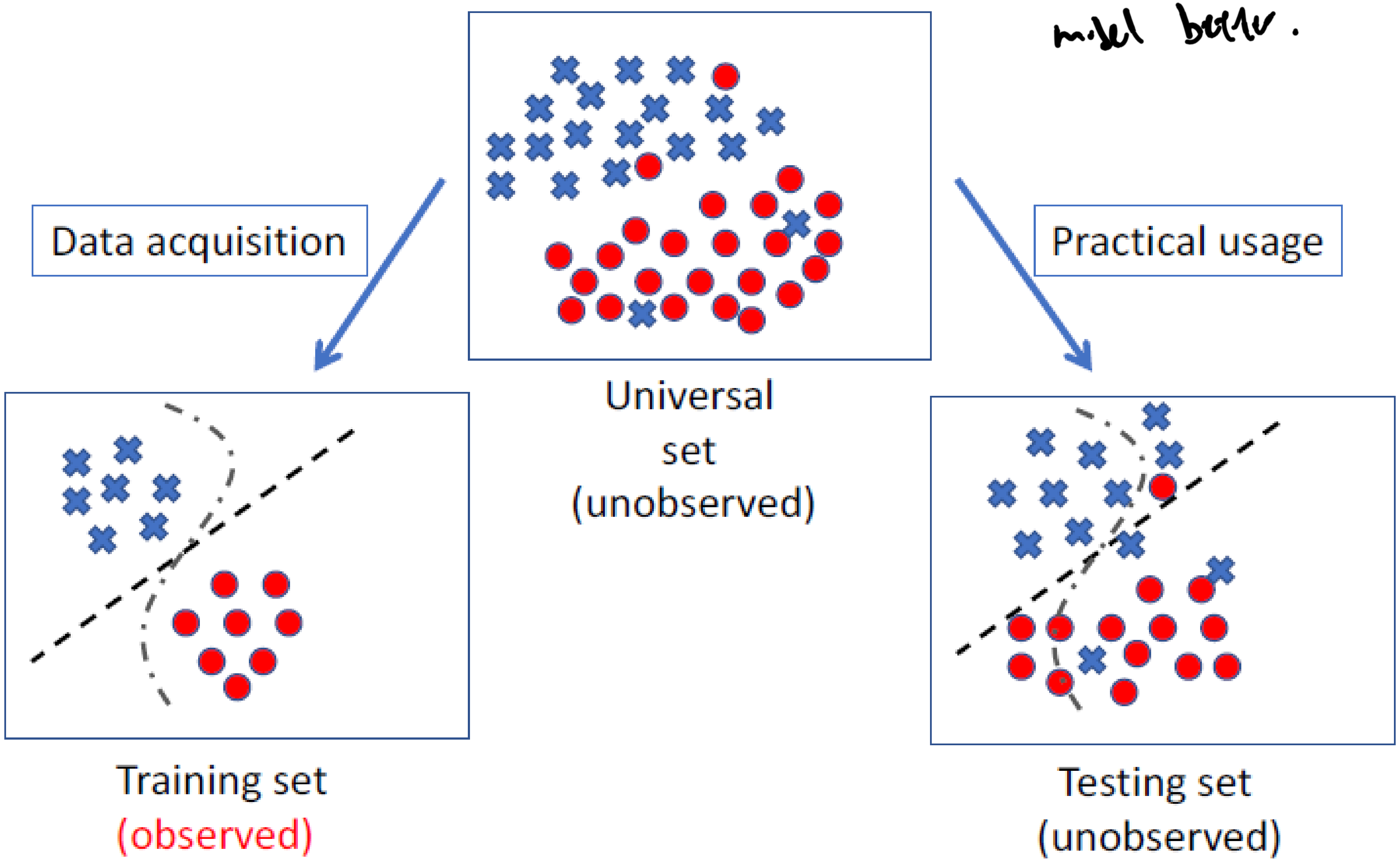
- Learning general models from a data of particular examples

- Data is cheap* and abundant (data warehouses, data mars); knowledge is expensive and scarce.

- Example in retail: Customer transactions to consumer behaviour:
  - *People who bought "Da Vinci Code" also bought "The Five People You Meet in Heaven" (www.amazon.com)*

- Build a model that is *a good and useful approximation* to the data.

Training and Testing



Why does the straight line fit better than the S-shaped?

∵ The data fits the straight line model better.

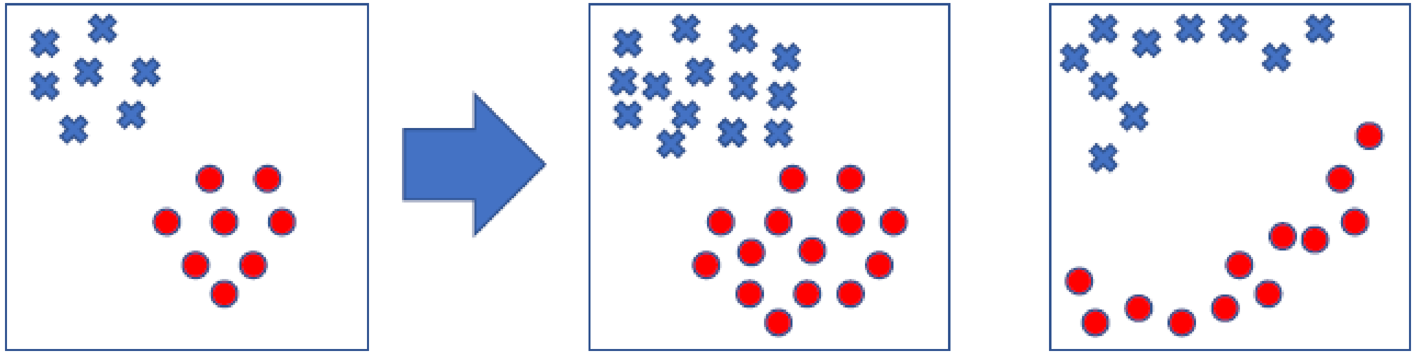
MAIN CONCEPT:

FITTING of MODEL.

- over fitting
- under fitting
  ↳ the model is not able to learn much & predict a useful outcome

Training and Testing

- Training is the process of making system able to learn.
- No free lunch rule:
  - Training set and testing set come from the same distribution
  - Need to make some assumption or bias



Training data

Testing data

Performance

- There are several factors affecting the performance:
  - **Quality of training data** provided → e.g. noisy data ⇒ bad quality
  - The form and extent of any initial **background knowledge**
  - The **type of feedback** provided
  - The **learning algorithm** used

- Two important factors:
  - Modelling → mapping of inputs → outputs
  - Optimization → making sure that the algorithm runs efficiently → e.g. time / space / data complexity

If your perfect model is not computationally optimised (takes forever to learn) is not optimizable & feasible

Algorithms

- The success of machine learning system also depends on the algorithms.

- The algorithms control the **search** to find and build the knowledge structures.

- The learning algorithms should **extract** useful information from training examples.

Types of Machine Learning

Based on information available

- **Supervised learning (**$\{x_n \in R^d, y_n \in R\}_{n=1}^N$**)**
  - Classification (discrete labels)
  - Regression (real values)
- **Unsupervised learning (** $\{x_n \in R^d\}_{n=1}^N$ **)**
  - Clustering
  - Probability distribution estimation
  - Finding association (in features)
  - Dimension reduction
- **Semi-supervised learning**

- combination of supervised & unsupervised

- **Reinforcement learning**
  - Decision making (robotics, board games)

- If the decision made is correct, you approve ⇒ letting the model know it is correct.

Based on learner's role

- **Passive learning**
- **Active learning**

→ if the model is learning based on information given in batches

→ if the model is learning as it goes when information is provided real-time

↳ e.g. chat GPT 3.5 / 4.0

Types of Machine Learning

Based on learner's role

- Traditionally, learning algorithms have been **passive learners**, which take a given batch of data and process it to produce a hypothesis or model.

Data → Learner → Model

- **Active learners** are instead allowed to query the environment

– Ask questions

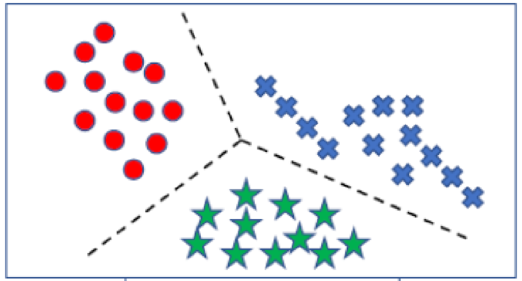
– Perform experiments

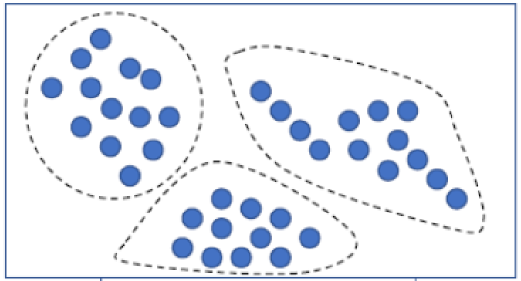
cost of experiment might be costly

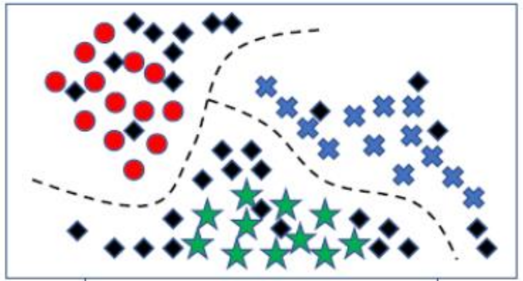
- **Open issues:** how to query the environment optimally? how to account for the cost of queries?

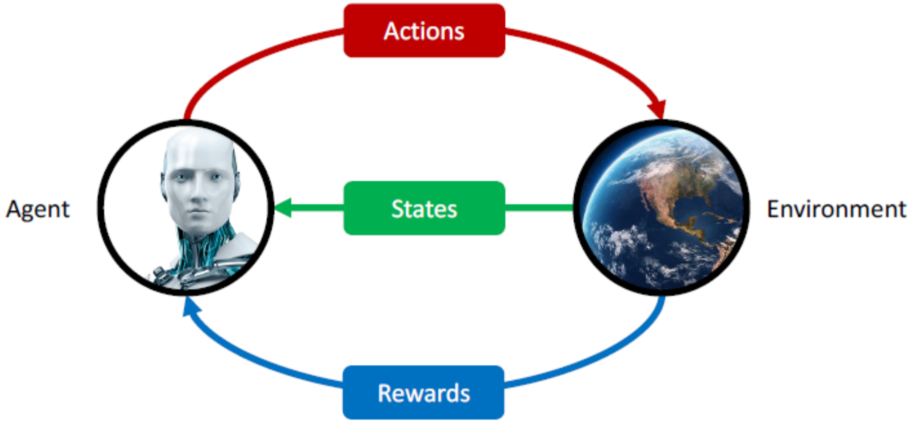
Types of Machine Learning

Based on information available


Supervised learning


Unsupervised learning


Semi-supervised learning



Reinforcement learning

Types of Machine Learning

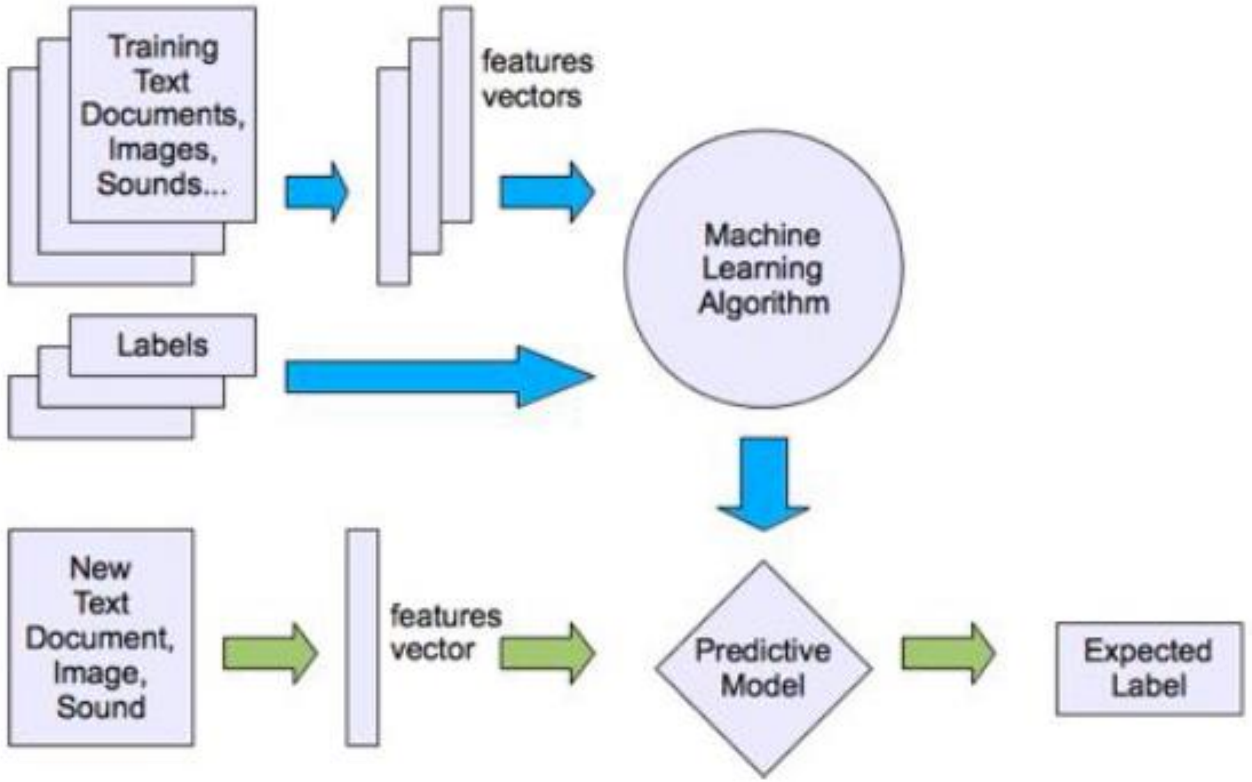
- **Supervised Learning**

Types of Machine Learning

- **Supervised Learning**

training phase

test phase

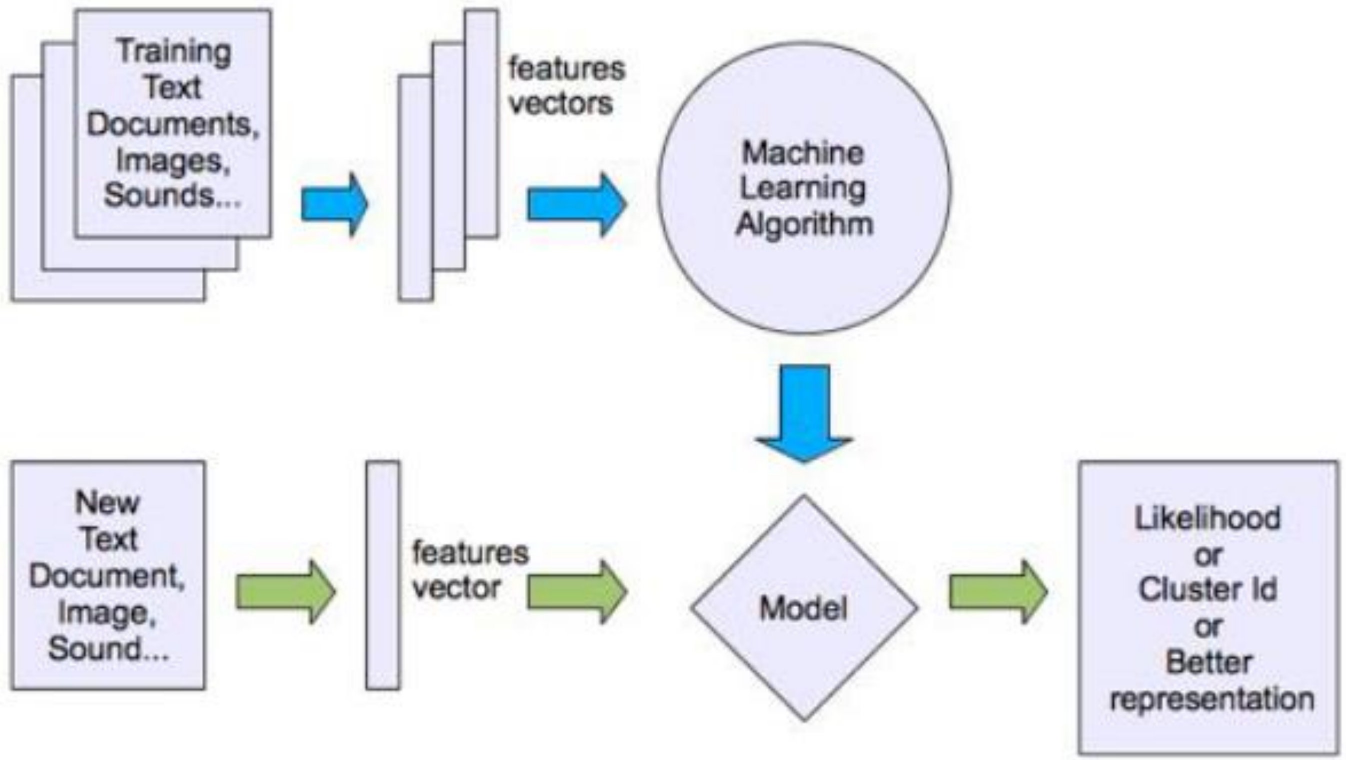


Types of Machine Learning

- **Unsupervised Learning**

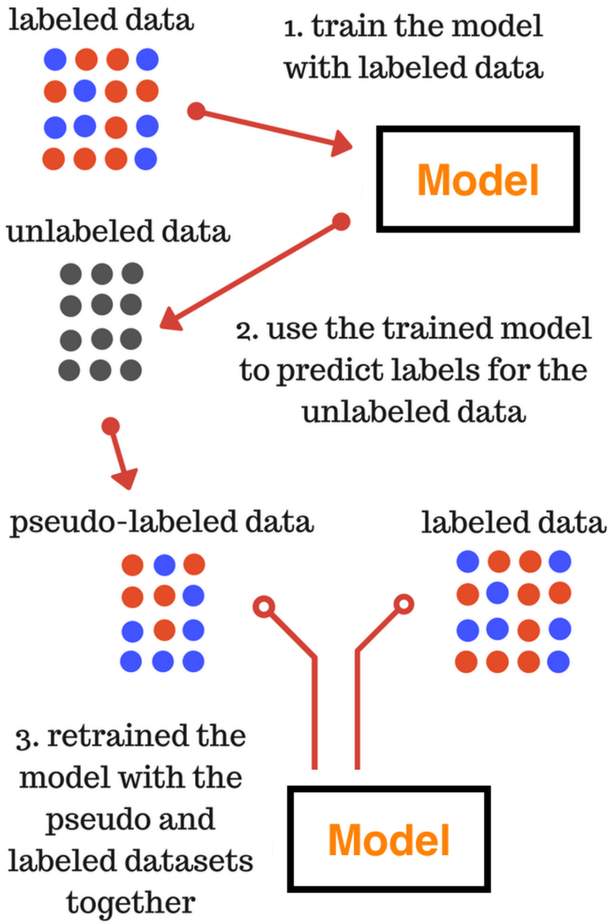
Types of Machine Learning

- **Unsupervised Learning**



Types of Machine Learning

- **Semi-supervised Learning**

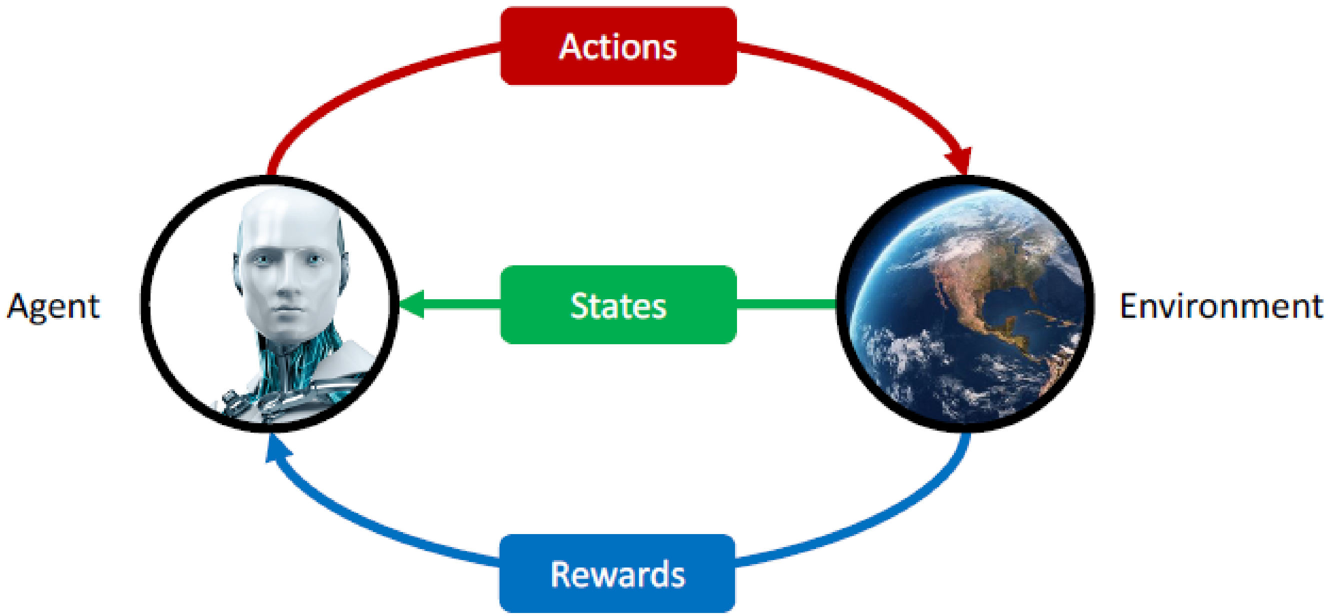


Types of Machine Learning

- **Reinforcement Learning:** Rewards from a sequence of actions

Types of Machine Learning

- **Reinforcement Learning**



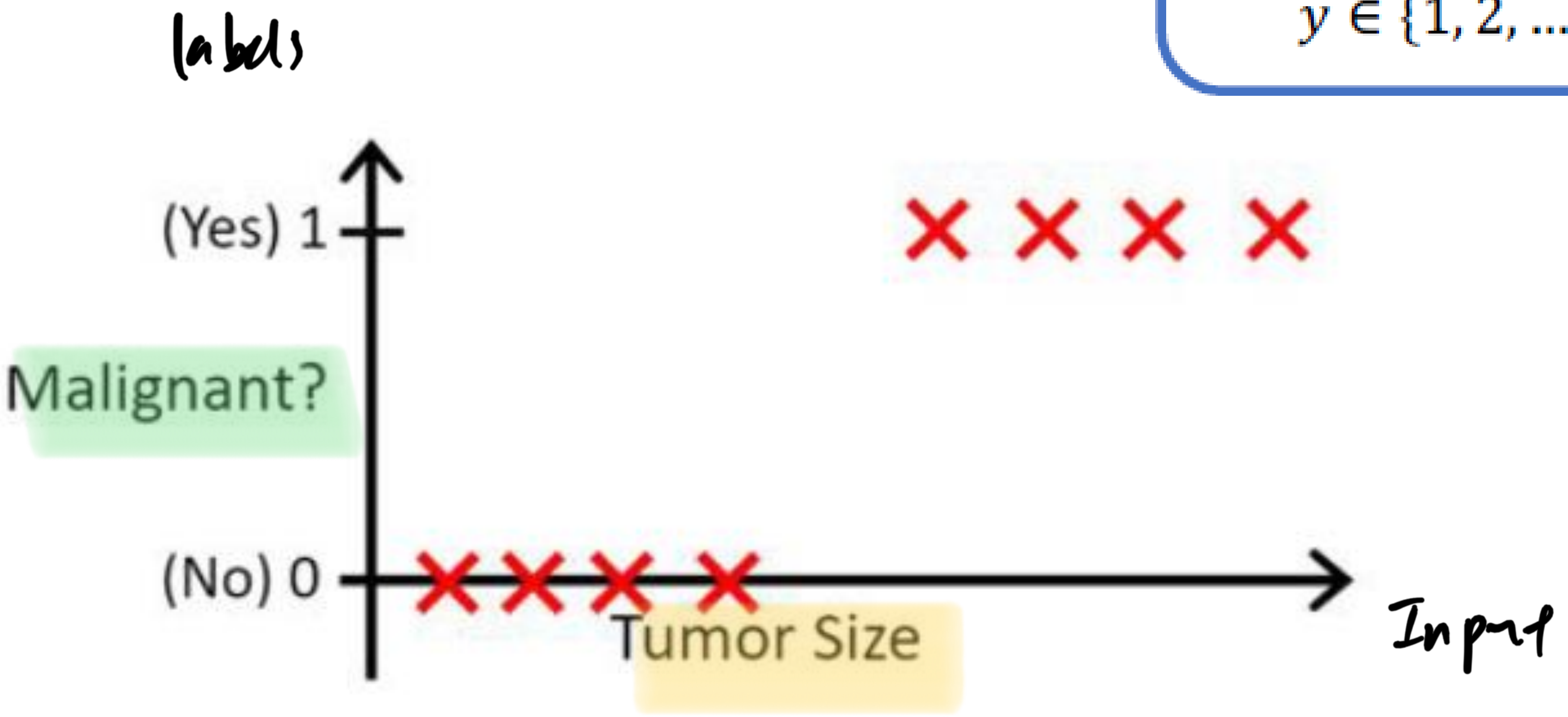
The state space can be discrete or continuous. In case of continuous states, you would use a function approximator to represent your state.

Learning Problems

Supervised Learning

- **Classification (1-d features)**

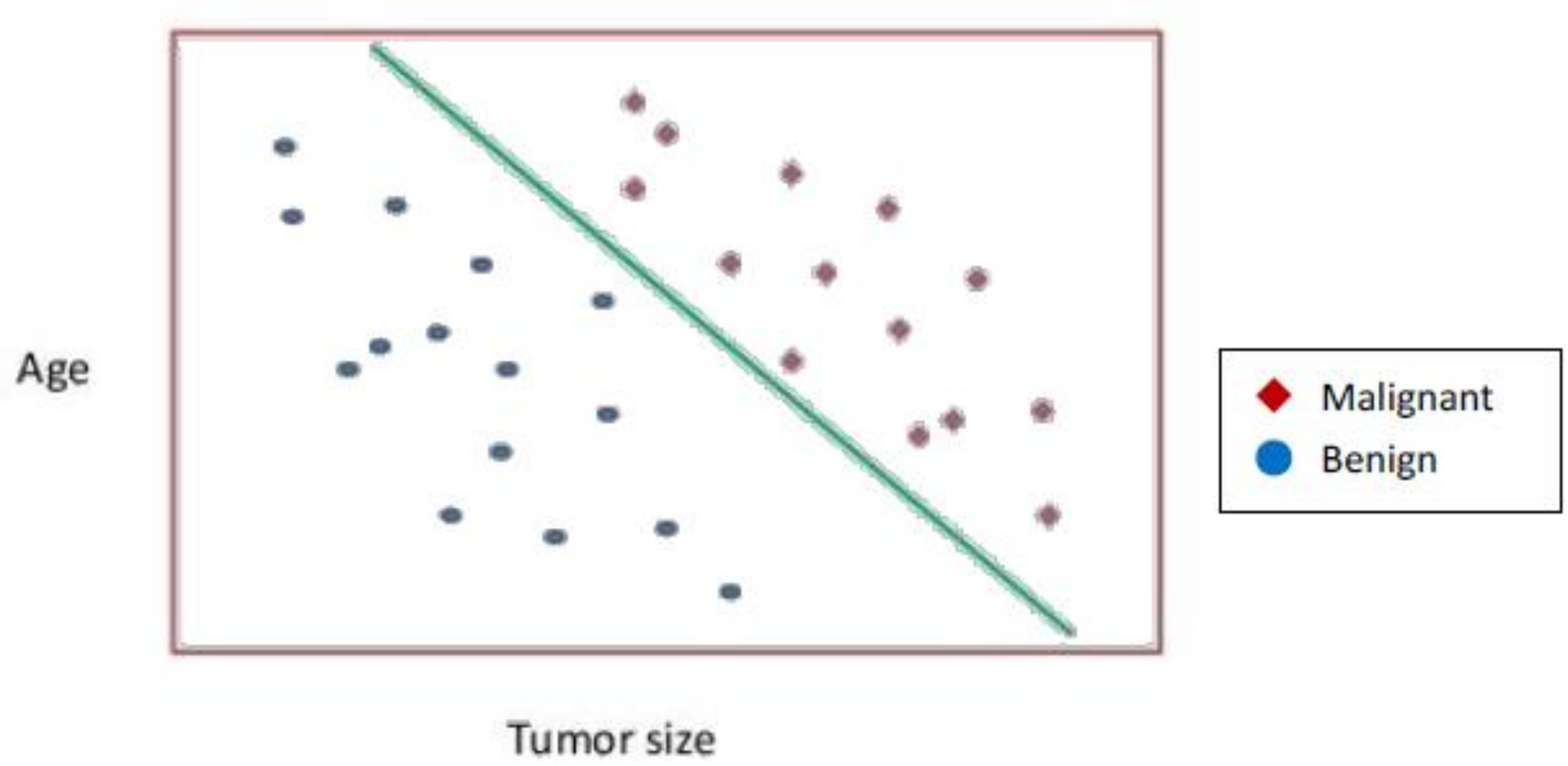
Learning a function
$$y = f(x)$$
$$x \in \mathbb{R}$$
$$y \in \{1, 2, \dots, k\}$$



Supervised Learning

- **Classification (2-d features) - Linear**

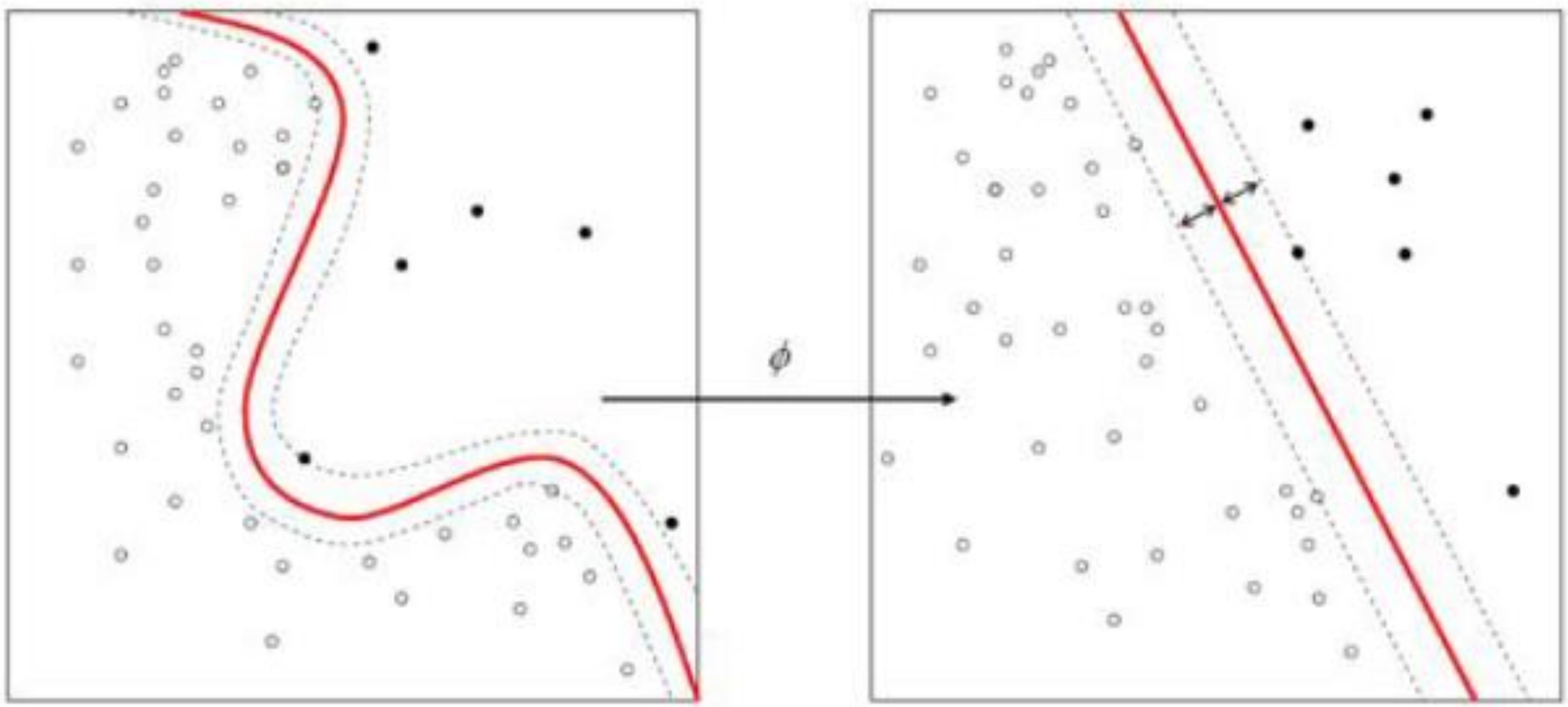
Inputs:
① Tumor size
② Age



Supervised Learning

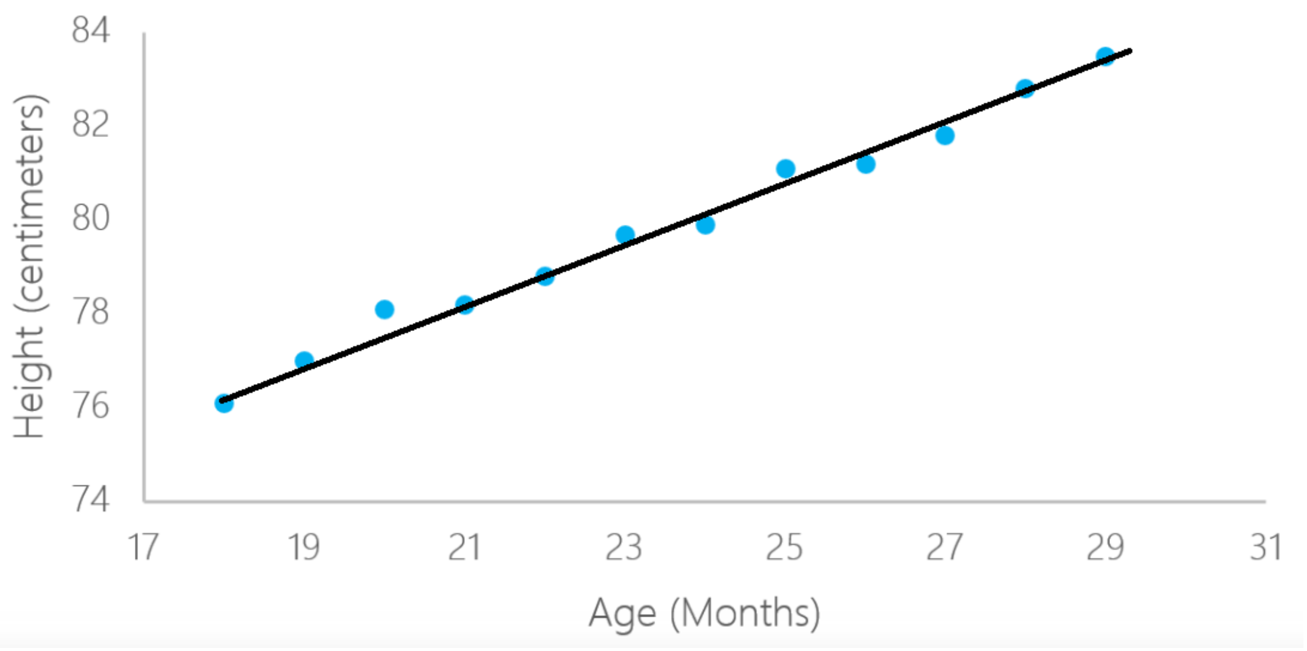
× WEEK 4

- **Classification (Non-Linear)**



Supervised Learning

- **Regression (Linear)**



Learning a function

$y = f(x)$

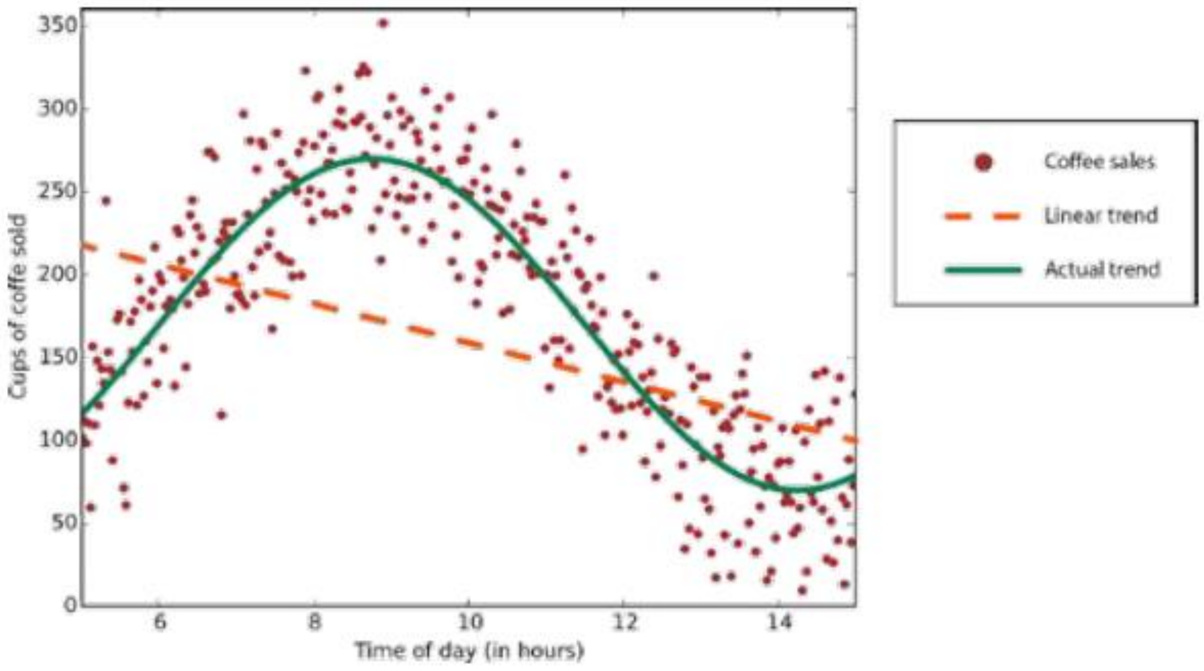
$x \in \mathbb{R}$

$y \in \mathbb{R}$

y is no longer a categorical data since it is supervised learning.

Supervised Learning

- **Regression (Non-Linear)**

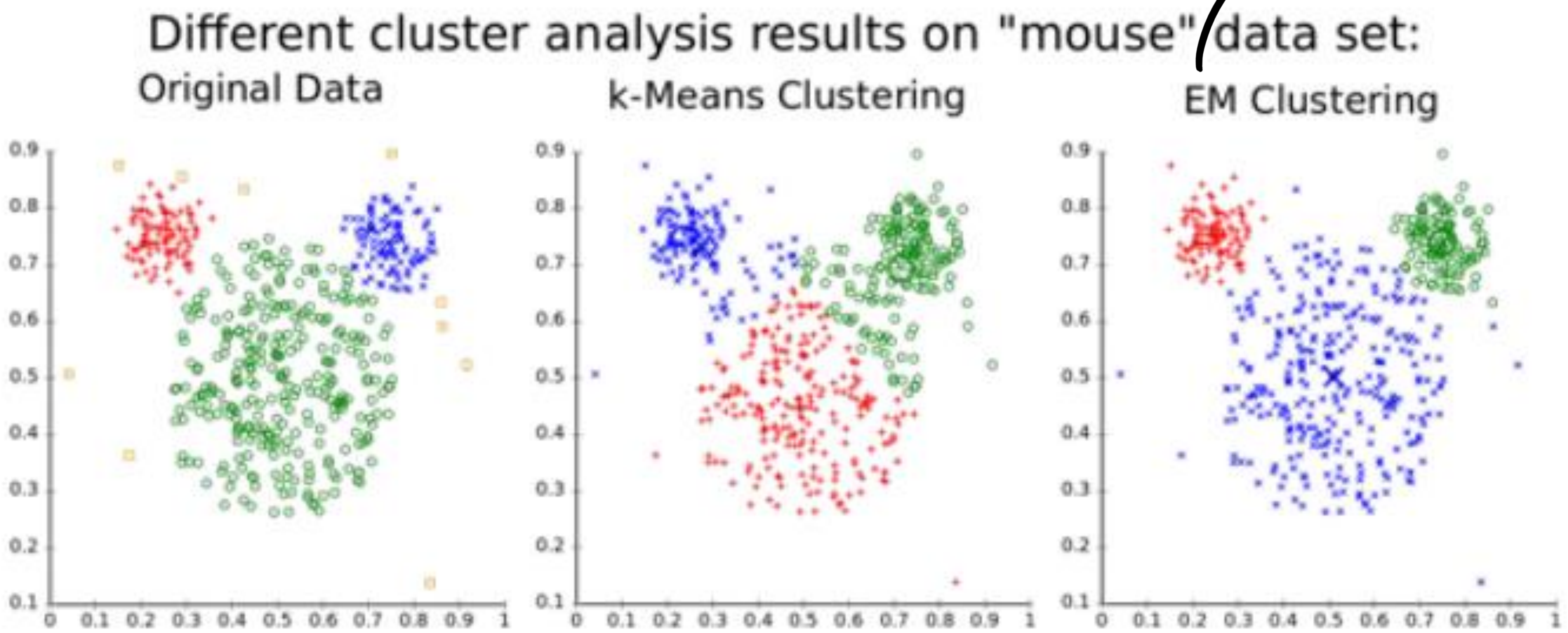


e.g. clustering genres of books based on content
annomaily detection – credit card fraud

Unsupervised Learning

expected-maximisation

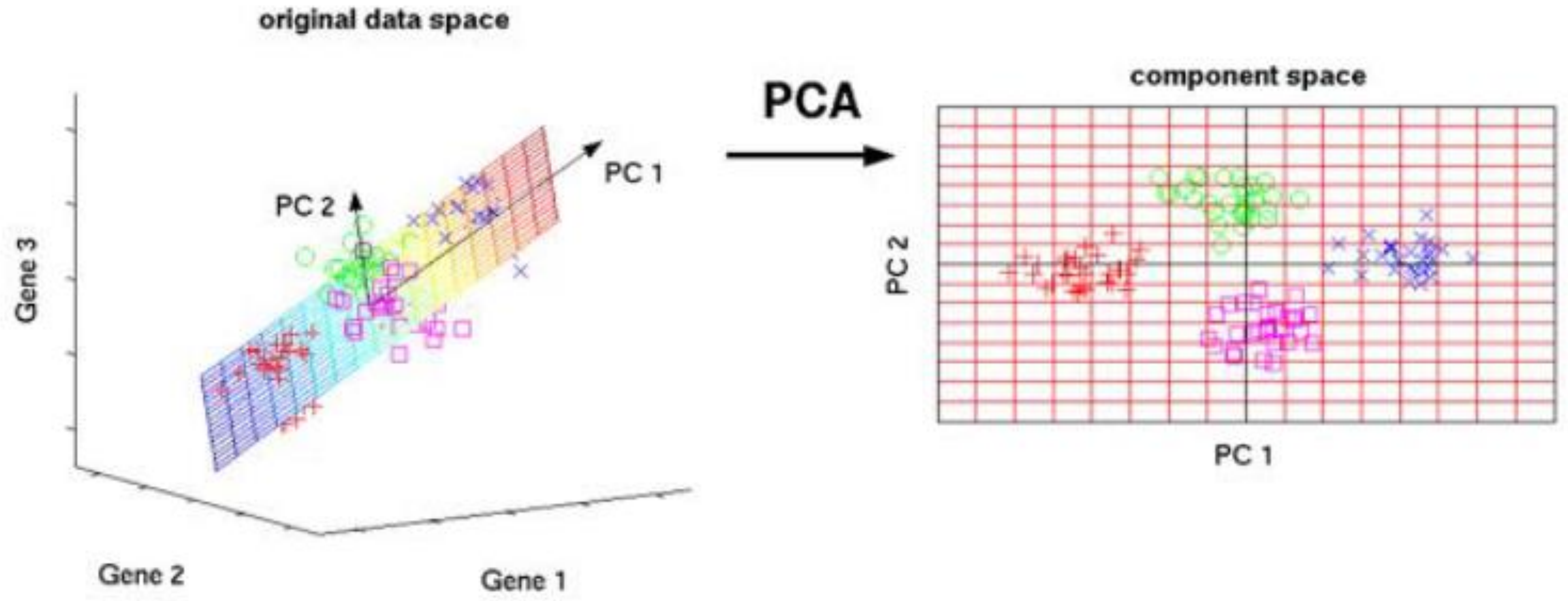
- **Clustering**



Unsupervised Learning

learning the things that is only important & contributes to a result

- **Dimensionality Reduction: Sub-space Learning**

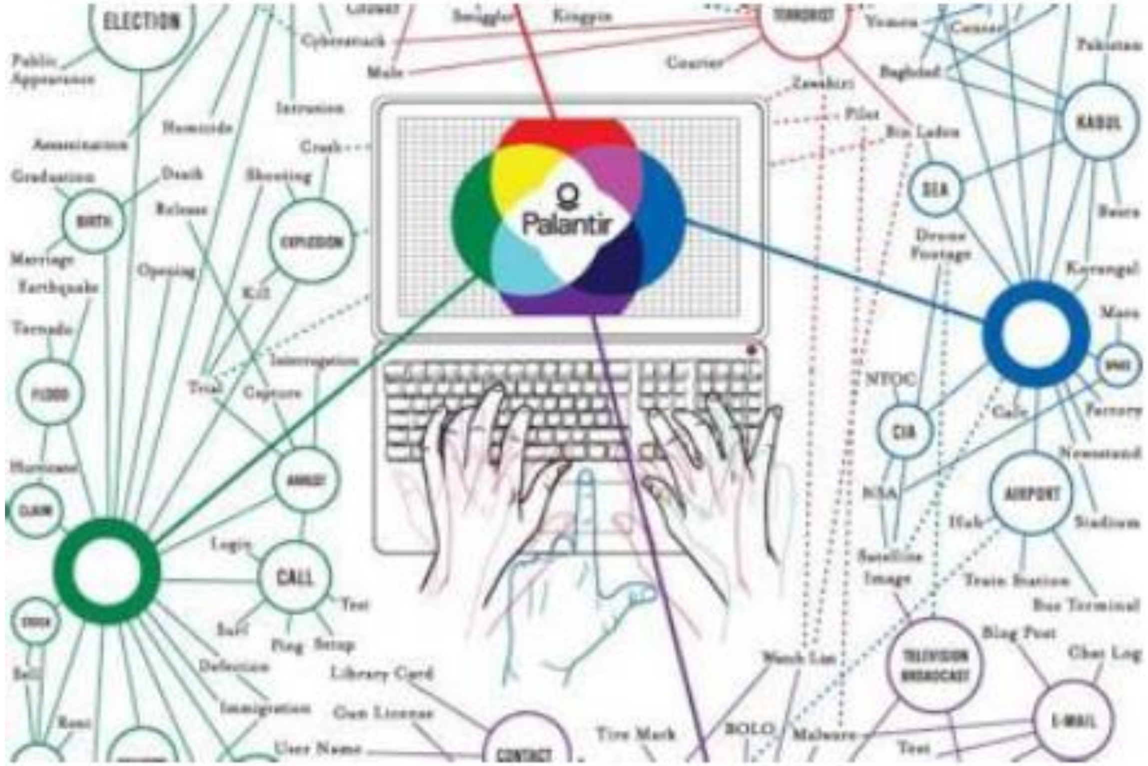


Applications

Fraud detection

PayPal wanted to pit fraud experts against the scammers. Engineers designed a software program that let human fraud experts to quickly sift through transaction data and start asking questions to raise a red flag. What type of machine learning is this?

supervised learning.
Since the expert is giving
the questions to raise red flags.



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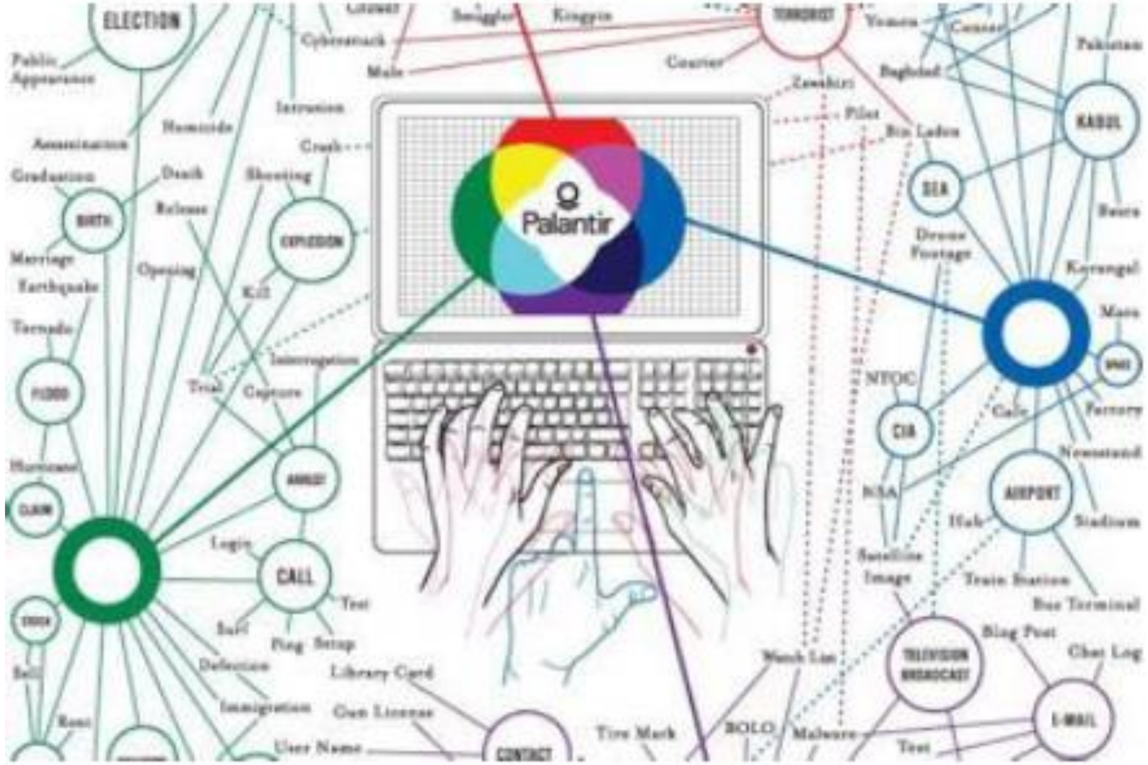
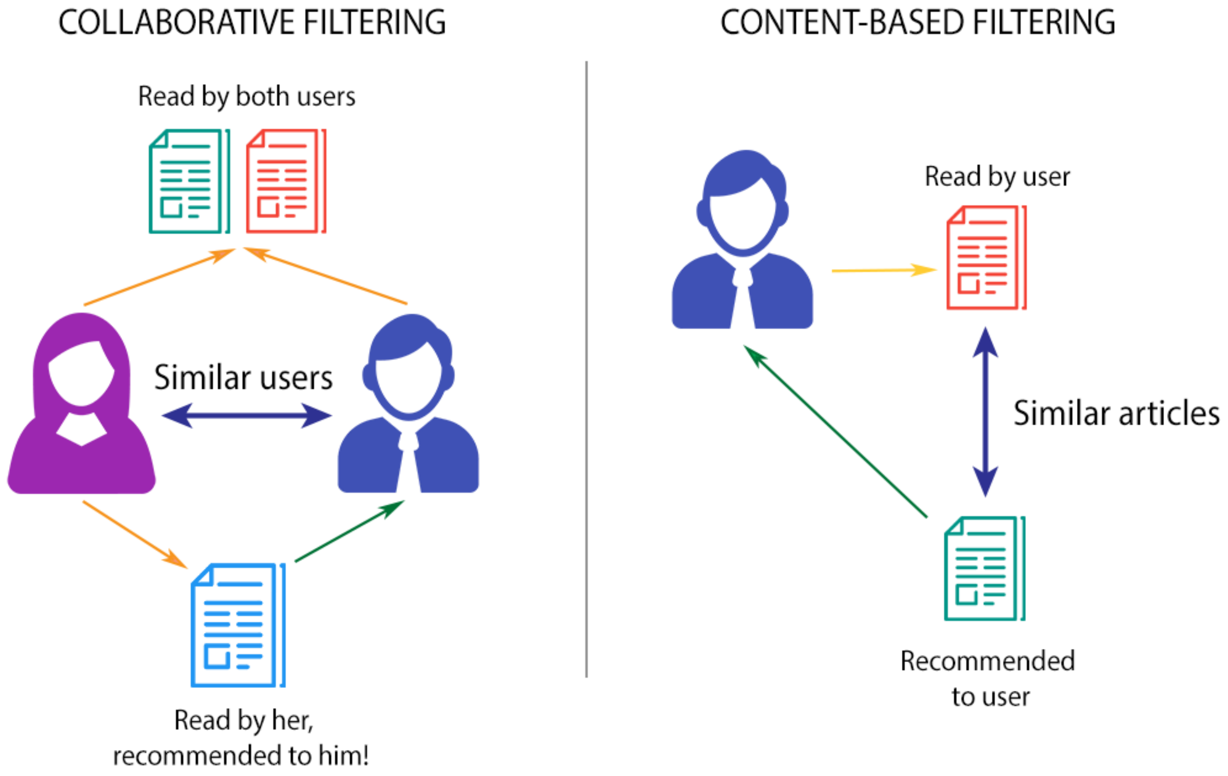


Supervised learning

https://www.washingtonian.com/2012/01/31/killer-app/

Recommender Systems

Recommendation engines filter out the products that a particular customer would be interested in or would buy based on their previous buying history for the existing customers and based on the preferred interests for new customers. What type of machine learning is this?



unsupervised.
There are no labels provided, it is based on each customer's previous buying history.
There is nobody to supervise the learning.

Recommender Systems

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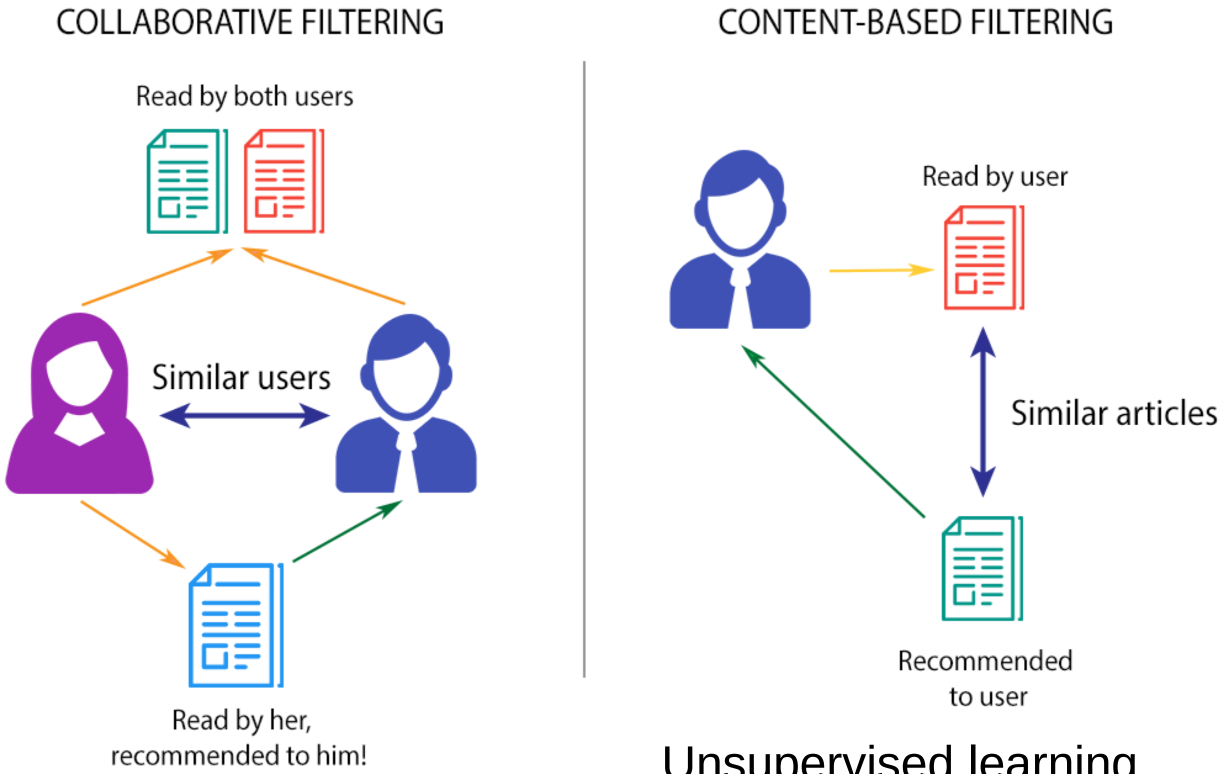
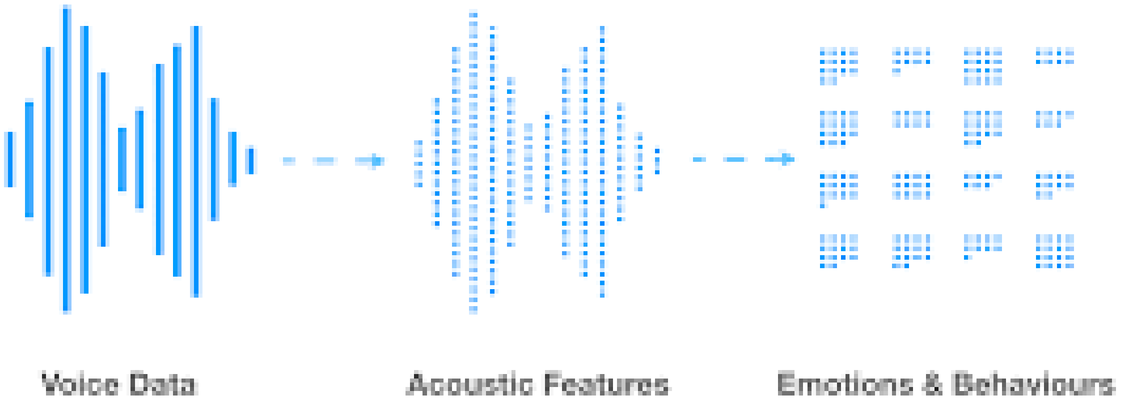


Unsupervised learning

https://towardsdatascience.com/brief-on-recommender-systems-b86a1068a4dd

Speech Analysis

Using small amount of diverse user data (age, gender, accent), we have labelled emotional behaviours and with that we would like to derived labels for a larger pool of users. What type of machine learning is this?



Semi-Supervised.

It consists of a combination of labels & non-labels.

Speech Analysis

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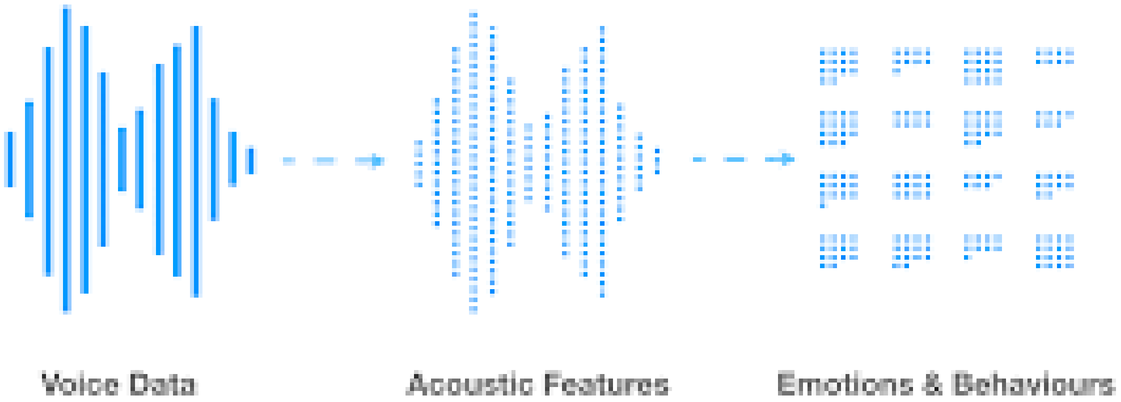


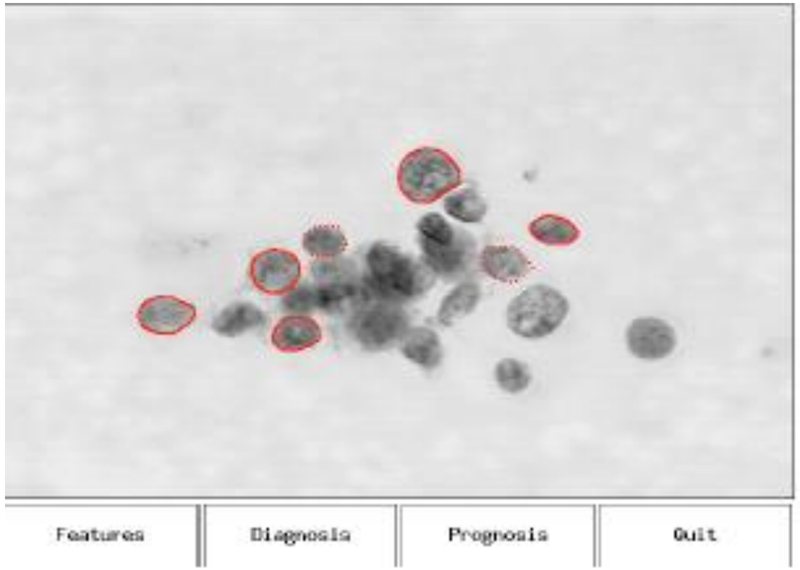
Semi-Supervised learning

https://en.wikipedia.org/wiki/Semi-supervised_learning

A case study

Supervised Learning

Example: A dataset



- Cell samples were taken from tumors in breast cancer patients before surgery, and imaged
- Tumors were excised

binary classification

- Patients were followed to determine whether or not the cancer recurred, and how long until recurrence or disease free

Regression → Predictions $\in \mathbb{R}$

Example: A dataset

features / attributes

- 30 real-valued variables per tumour → e.g. size, texture, perimeter
- Two variables that can be predicted:
  - Outcome (R = recurrent, N = non-recurrent)
  - Time (until recurrence, for R, time healthy for N).

1 tumour

tumor size	texture	perimeter	. . .	outcome	time
18.02	27.6	117.5		N	31
17.99	10.38	122.8		N	61
20.29	14.34	135.1		R	27

Terminology

tumor size	texture	perimeter	...	outcome	time
18.02	27.6	117.5		N	31
17.99	10.38	122.8		N	61
20.29	14.34	135.1		R	27

- Columns are called *input variables* or *features* or *attributes*
- The outcome and time (which we are trying to predict) are called *output variables* or *targets or responses*.
- A row in the table is called *training example* or *instance*
- The whole table is called (*training) data set*.
- The problem of predicting the recurrence is called *(binary) classification*.
- The problem of predicting the time is called *regression*.

More formally

- Let $\mathcal{X}$ denote the space of input values
- Let $\mathcal{Y}$ denote the space of output values
- Given a data set $\mathcal{S}_n \subset \mathcal{X} \times \mathcal{Y}$, find a function:

$$h : \mathcal{X} \to \mathcal{Y}$$

  such that $h(\mathbf{x})$ is a *"good predictor"* for the value of y.
- h is called a *classifier*
- Problems are categorized by the type of output domain
  - If $\mathcal{Y} = \mathbb{R}$, this problem is called *regression*
  - If $\mathcal{Y}$ is a categorical variable (i.e., part of a finite discrete set), the problem is called *classification*
  - If $\mathcal{Y}$ is a more complex structure (eg graph) the problem is called *structured prediction*

More formally

tumor size	texture	perimeter	...	outcome	time
18.02	27.6	117.5		N	31
17.99	10.38	122.8		N	61
20.29	14.34	135.1		R	27

Training data

$$\mathcal{S}_n = \{ (x^{(i)}, y^{(i)}) \mid i = 1, \dots, n \}$$

Index of the row

- Features/Inputs $x^{(i)} = \left(x_1^{(i)}, \dots, x_d^{(i)}\right)^{\mathsf{T}} \in \mathbb{R}^d$
- Response/Output $y^{(i)} \in \mathbb{R}$ or $y \in \{1, 2, \dots, k\}$

dimensionality of variable

↳ e.g. 30 features
$d = \{1, \dots, 30\}$

Steps to solving a supervised learning problem

e.g. encode into 1 or 0
Y or N

useful for multi-class classification

1. Decide what the input-output pairs are.
2. Decide how to encode inputs and outputs.
   - This defines the input space X and the output space Y.
3. Choose a **hypothesis class H (model)**.
4. Choose an **error function** (**cost function**) to define the best hypothesis.
5. Choose an **algorithm for searching** efficiently through the space of hypotheses (**optimizing**).

Key aspects of learning problems

- **Set of classifiers H:** modelling
  - Different settings of parameters give different classifiers in the set

- **Learning algorithm / Criterion:** optimizing

- **Generalization**
  - Choice of H (too big → **overfit**, too small → **underfit**)
  - Training data S_n
  - Learning algorithm

Intended Learning Outcomes

- Define machine learning in terms of **algorithms**, **tasks**, **performance** and **experience**.

- List four main types of machine learning, e.g. **supervised, unsupervised, reinforcement learning, semi-supervised learning**

- Describe **essential components** required for building a machine learning model: training data, hypothesis class and learning algorithm.

Linear classification

Linear Classification

- Linear classifier is a particular constrained set of classifiers

weight of feature

$$h(x;\theta) = \text{sign}(\theta_1 x_1 + \ldots + \theta_d x_d) = \text{sign}(\theta \cdot x) = \begin{cases} +1, & \theta \cdot x \geq 0 \\ -1, & \theta \cdot x < 0 \end{cases}$$

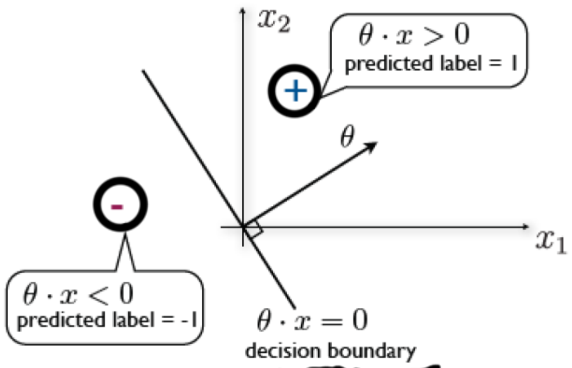
- where, $\theta \cdot x = \theta^T x$ and $\theta = [\theta_1, \ldots, \theta_d]^T$ is a column vector of real valued parameters or weights

$$\theta^T x = [\theta_1 \cdots \theta_d] \begin{bmatrix} x_1 & & x_n \\ \vdots & & \vdots \\ x_d & & x_d \end{bmatrix} [y_1 \ldots y_n] = y$$

weights $1 \times d$

feature $d \times n$

$1 \times n$



decision boundary passes through the origin

$\therefore \theta \cdot x = 0$

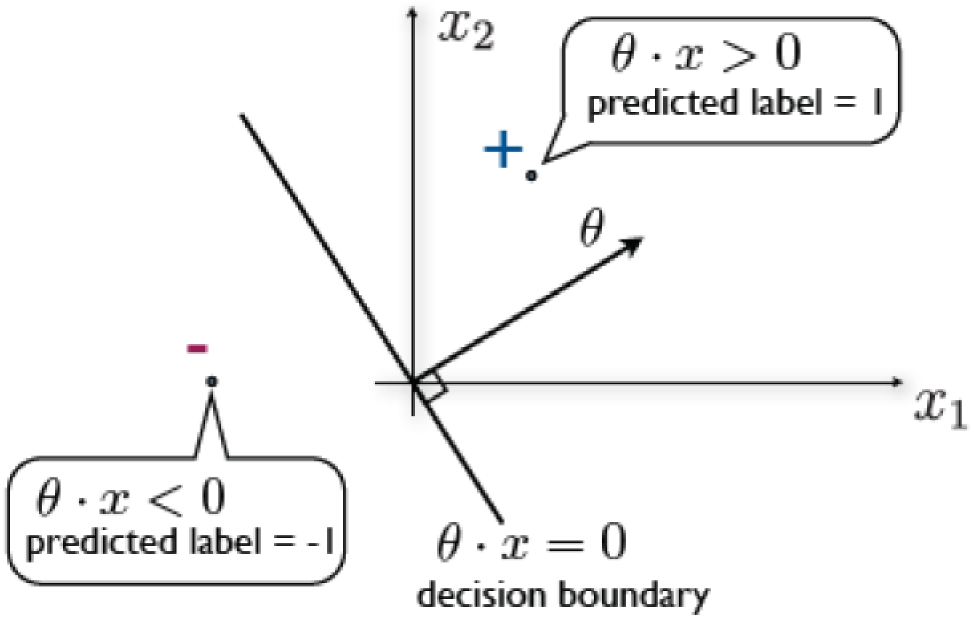
Goal: To find the decision boundary so to classify the output

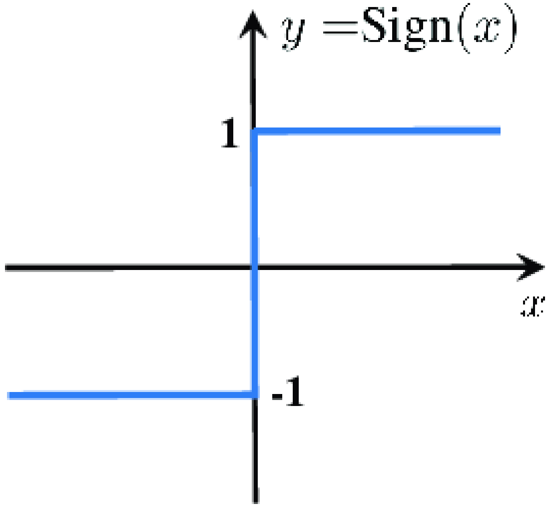
$\theta \cdot x$ will have label +1.

Linear Classification

everything that lies on the decision boundary will have label +1

$$h(x;\theta) = \text{sign}(\theta_1 x_1 + \ldots + \theta_d x_d) = \text{sign}(\theta \cdot x) = \begin{cases} +1, & \theta \cdot x \geq 0 \\ -1, & \theta \cdot x < 0 \end{cases}$$





the options of the decision boundary

⇒ the slope → angle of the decision

⇒ the y-intercept → offset of the decision.